General Sir John Kotelawala Defence University

Faculty of Management, Social Sciences, and Humanities Department of Languages BSc in Applied Data Science Communication

1st Year: Semester
2 SQL for data science- (LB 1224)
Assignment 1
Lecture: Dr. Charith Silva

Team Members- KiruliGamage(D/ADC/24/0011)
Binath Kumarasinghe (D/ADC/24/0016)
Pathum Athukorala (D/ADC/24/0037)

# Global Olympic Performance Analysis

## Introduction

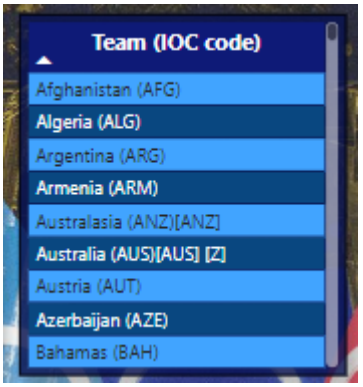
Held every four years, the Olympic Games are an international multi-sport competition including competitors from many athletic disciplines. The top three finishers in each event get medals (gold, silver, and bronze), which stand for excellence and accomplishment in athletics.

An Olympic medal is a noteworthy achievement that is the result of years of commitment and labour. Nations frequently take great pride in the accomplishments of its athletes, and the number of medals won is carefully followed as an indicator of success. An Olympic medal is a remarkable accomplishment that takes years of dedication and hard work to attain. Athletes' achievements are often seen with great pride by their respective nations, and the quantity of medals won is closely monitored as a sign of success. Smaller nations can still achieve great success at the Olympics, even if highly competitive nations with strong sports cultures, typically lead in medal counts.

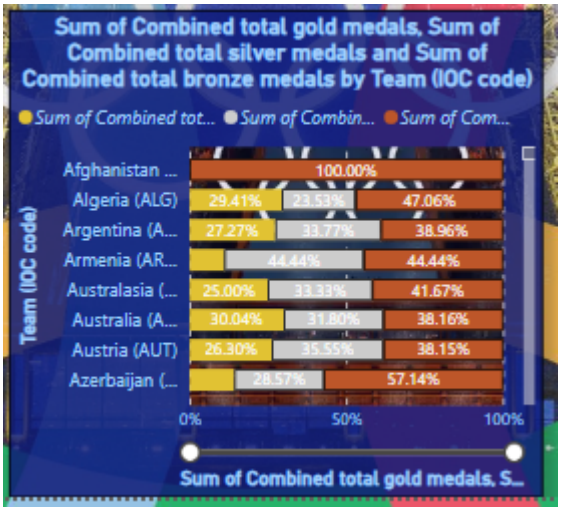
In this Power Bi dashboard we can see the data and the analysis of the previous winter and summer olympics and the relationships and insights and trends .This report presents an analysis of the Olympic Games data, focusing on key performance metrics, geographical distribution, athlete performance, and medal distribution. The visualizations provide insights

into the trends and patterns observed in various Olympic events, highlighting the standout performances, dominant countries, and overall outcomes.

## Patterns Observed




The first table shows the countries that previously participated in the winter and summer olympics . There are 156 countries who participated in the winter and summer olympics . Its a table that represents all the medal counts of each country from 1896 to 2022.

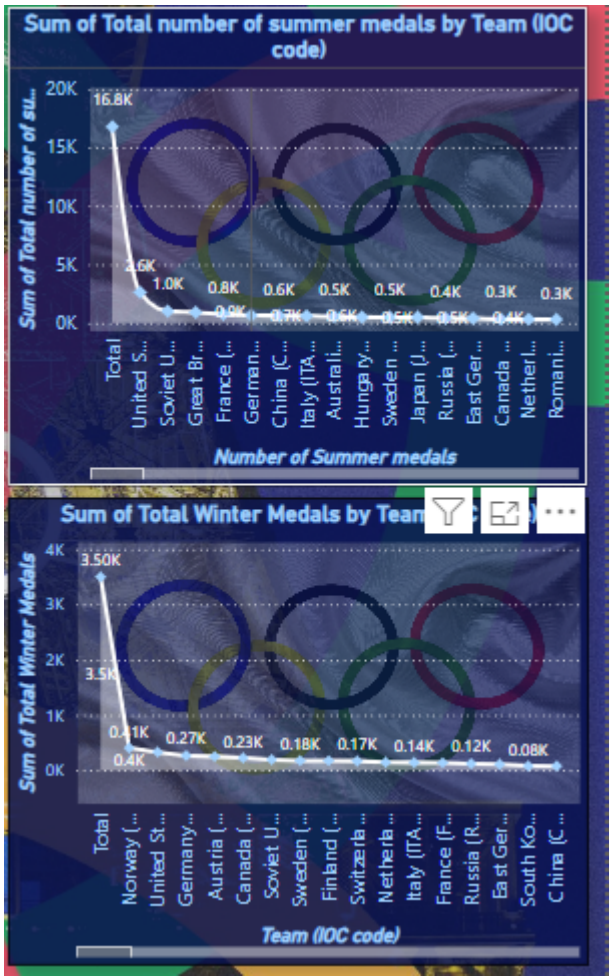



In this visualisation from the clustered bar chart we can see the representation of the sum of the combined amount of gold medals ,silver medals and bronze medals,

Observations

- United states is dominating with the most number of medals including gold ,silver and bronze

- Their is also some rising nations such as China and Russia which is emerging with a notes number of gold , silver and bronze medals




In these clustered bar charts it shows the total number of  Summer medals and the the total number of winter medals.

Observations

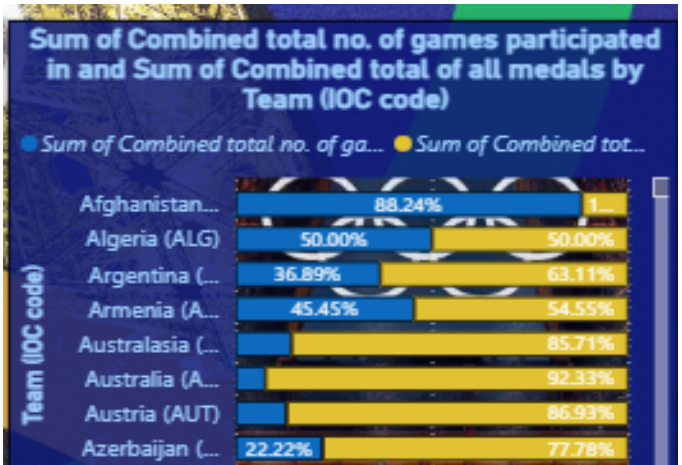
- Norway has the most number of winter medals with the sum of 405 medals

- United States has the most number of winter models with the sum od 2629 medals

From the map above we can see the total number of medals that had been won by each country . The map only represents the top 10 countries that have the most number of total medals . The bubble shows the amount of the medals won by the country .

Observations

- We can clearly see from the map above that the United States of America has the most medals .



The Clustered Bar Chart shows the sum of combined total number of games participated and the total medals won by each team .

Observations

- United States Of America has won 94% medals from the games that they participated in which were the highest .

- Sri Lanka has only won 1 medal which is 10% from the 90% Games that they participated in ..

**<u>Key Observations</u>**

**1.Top Countries by Medal Count**

The visualisations reveal that certain countries consistently top the medal counts, indicating their dominance in the Olympic Games. These countries often have strong athletic programs, significant investment in sports, and a history of Olympic success. The data likely shows a concentration of medals in traditional sports powerhouses like the United States, China, and Russia. These countries' continued success underscores the importance of robust training programs and governmental support for sports.

**2.Medal Distribution Over Time**

The time-series analysis of medal distribution shows trends in how certain countries or sports have performed over multiple Olympic Games. For example, a country may have a period of dominance in a specific sport that correlates with increased investment or the emergence of star athletes. Like USA who has the most number of medals .

**Global Medal Distribution**

The map visualisation of global medal distribution illustrates the geographical spread of Olympic success. Regions with a higher concentration of medals are like America and Europe has likely better sports infrastructure, training facilities, and investment in athletes. The data shows a clear division between developed and developing nations, with the former generally achieving more Olympic success. However, emerging nations might also be highlighted, showcasing their progress in sports on the global stage.

**3.Regional Performance in Specific Sports**

Certain regions may excel in specific sports, influenced by cultural factors, climate, and available resources. For example, Nordic countries have performed better in winter sports. Understanding these regional strengths allows for a more nuanced view of global athletic performance and can inform future investments and training programs.

**4.Continent-Wise Medal Distribution**

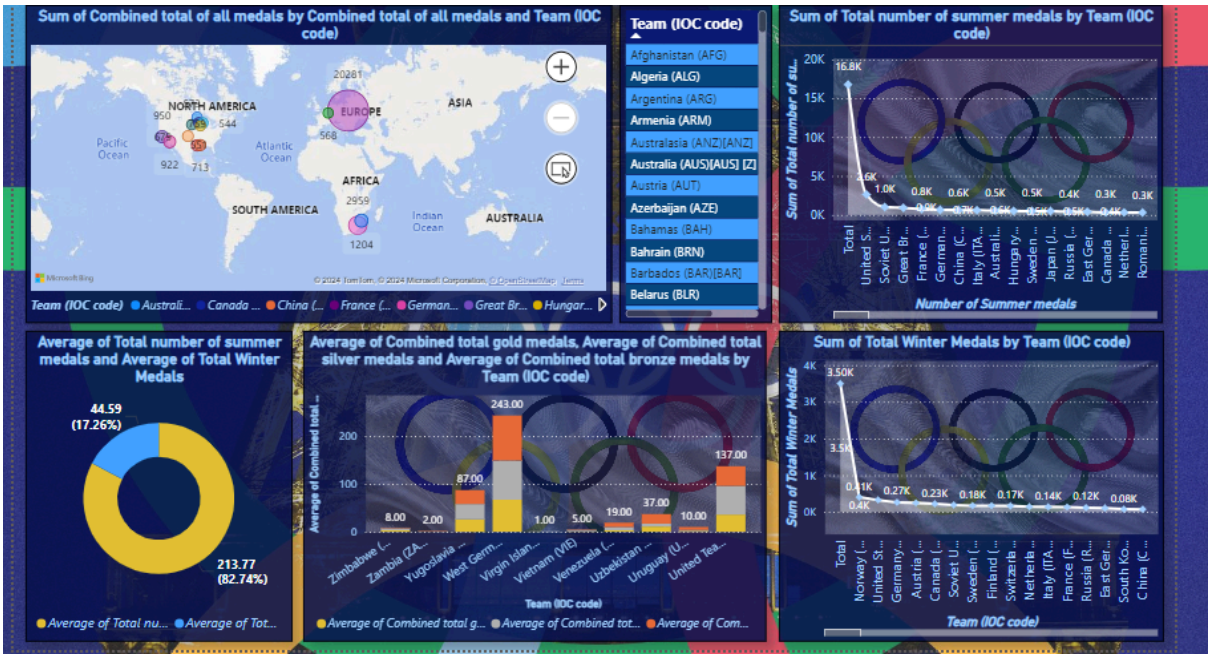
The continent-wise analysis of medal distribution might highlight disparities between different regions. Continents like Europe, North America, and Asia are likely to dominate the medal counts, reflecting their economic power, investment in sports, and large populations. The data could also show the growing influence of other continents, such as South America and Africa, in certain sports, indicating a more global spread of Olympic success.

## Conclusion

This report provides a comprehensive analysis of the global Olympic performance, highlighting key trends, regional strengths, and athlete achievements. The data reveals clear patterns of dominance by certain countries and sports, as well as emerging opportunities for others.From This analysis we can see all the countries performances and insights of the data,

# Dashboard

Global Olympic Performance Analysis
Sum of Combined total gold medals, Sum of Combined total silver medals and Sum of Combined total bronze medals
13K (32.7%)
13K (32.53%)
14K (34.77%)
Sum of Combi...
Sum of Combi...
Sum of Combi...
Team (IOC code)
Afghanistan (AFG)
Algeria (ALG)
Argentina (ARG)
Armenia (ARM)
Australasia (ANZ)[ANZ]
Australia (AUS)[AUS] [Z]
Austria (AUT)
Azerbaijan (AZE)
Bahamas (BAH)
157
Count of Team (IOC code)
13K
Sum of Combined total gold medals
41K
Sum of Combined total of all med...
Sum of Total number of summer medals and Sum of Total Winter Medals by Team (IOC code)
Sum of Total number of summer ...
Sum of Total Winter ...
Team (IOC code)
Sum of Total number of summer m...
Sum of Combined total gold medals, Sum of Combined total silver medals and Sum of Combined total bronze medals by Team (IOC code)
Sum of Combined tot...
Sum of Combin...
Sum of Com...
Sum of Combined total gold medals, S...
Sum of Combined total no. of games participated in and Sum of Combined total of all medals by Team (IOC code)
Sum of Combined total no. of ga...
Sum of Combined tot...
Sum of Combined total no. of gam...


Sum of Combined total of all medals by Combined total of all medals and Team (IOC code)
NORTH AMERICA
EUROPE
ASIA
AFRICA
SOUTH AMERICA
AUSTRALIA
Pacific Ocean
Atlantic Ocean
Indian Ocean
Team (IOC code)
Australi...
Canada ...
China (...
France (...
German...
Great Br...
Hungar...
Team (IOC code)
Afghanistan (AFG)
Algeria (ALG)
Argentina (ARG)
Armenia (ARM)
Australasia (ANZ)[ANZ]
Australia (AUS)[AUS] [Z]
Austria (AUT)
Azerbaijan (AZE)
Bahamas (BAH)
Bahrain (BRN)
Barbados (BAR)[BAR]
Belarus (BLR)
Sum of Total number of summer medals by Team (IOC code)
Sum of Total number of su...
Number of Summer medals
Average of Total number of summer medals and Average of Total Winter Medals
44.59 (17.26%)
213.77 (82.74%)
Average of Total nu...
Average of Tot...
Average of Combined total gold medals, Average of Combined total silver medals and Average of Combined total bronze medals by Team (IOC code)
Average of Combined total ...
Team (IOC code)
Average of Combined total g...
Average of Combined tot...
Average of Com...
Sum of Total Winter Medals by Team (IOC code)
Sum of Total Winter Medals
Team (IOC code)